12 The diagram shows a sector of a circle with centre O and radius 9 cm.

NOT TO SCALE

The perimeter of the sector is $(18 + 2\pi)$ cm.

Find the area of the sector.
Give your answer in terms of π.

$\frac{\theta}{360} 2\pi r + 18 = 18 + 2\pi$ $\qquad \theta = 40°$

$\frac{\theta}{360} \times 18\pi = 2\pi$ $\qquad \frac{40}{360} \times \pi 9^2$

$18\pi\theta = 2\pi \times 360$ $\qquad \frac{\pi \times 81}{9}$

$\frac{18\pi}{18\pi}\theta = \frac{720\pi}{18\pi}$ $\qquad 9\pi \text{ cm}^2$

9π cm² [4]

13 These are Rahul's 10 test scores.

9 8 9 10 7 x 9 9 x 7

The mean of these scores is 8.

Find the interquartile range.
You must show all your working.

$\frac{9 + 8 + 9 + 10 + 7 + 2x + 9 + 9 + 7}{10} = 8$

~~68 + 2x = 80~~

$68 + 2x = 80$

$2x = 80 - 68$

$\frac{2x}{2} = \frac{12}{2}$

$x = 6$

6, 6, 7, 7, 8, 9, 9, 9, 9, 10

$LQ = \frac{10 + 1}{4}$

$= 2.75^{th}$ place

$= \frac{6 + 7}{2} = 6.5$

$UQ = \frac{30 + 3}{4}$

$= 8.25^{th}$ place

$= \frac{18}{2}$

$= 9$

$IQ = 9 + 6.5$

$= 15.5$

15.5 [4]